

Physics-based Character Control Using conditional GAIL

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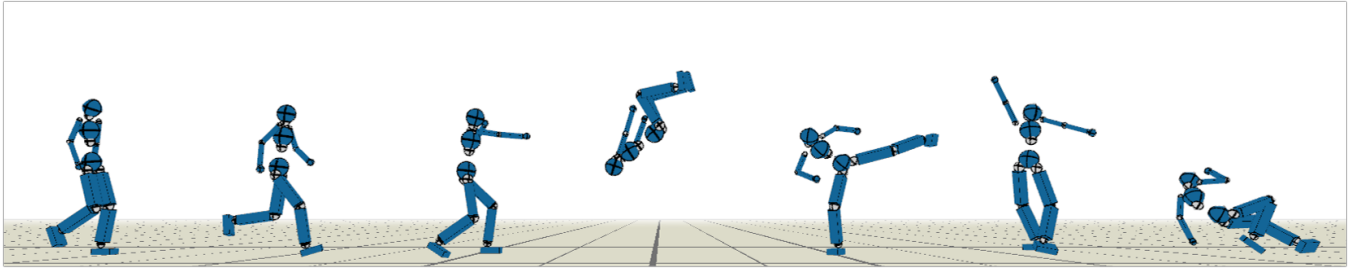


Figure 1: Our framework enables users to control physically simulated character that learns proper behaviors from multiple disparate motion clips following the user's input. This figure shows characters learn to do various actions following user control while walking.

ABSTRACT

The goal of our research is to control a physics-based character that learns several dynamic motor skills using conditional Generative adversarial imitation learning (GAIL). We present a network-based learning algorithm that learns various motor skills and changing motions between the motor skills from disparate motion clips. The overall framework for our controller is composed of a control policy which generates a character's behavior, and a discriminator which induces the policy to produce proper motions from a user's commands. The discriminator and the policy take outputs from each other as input and improve each performance through an adversarial training process. Using this system, when a user commands a specific motion to the character, the character can design a motion plan to perform the motion from the current pose. We demonstrated the effectiveness of our approach through examples with an interactive character that learns various dynamics motor skills and follows a user command in the physics simulation.

CCS CONCEPTS

• **Computing methodologies** → **Procedural animation; Physical simulation; Adversarial learning.**

KEYWORDS

character animation, physics-based character control, adversarial learning

ACM Reference Format:

Yongwoo Lee and Jehee Lee. 2022. Physics-based Character Control Using conditional GAIL. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Posters (SIGGRAPH '22 Posters)*, August 07-11, 2022. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3532719.3543224>

1 INTRODUCTION

Character control in physics simulation is one of the long-standing problems in computer graphics. From biped locomotion to dynamic motions, physics-based character control has been improved by various algorithms.

Recently, data-driven approaches using deep reinforcement learning (RL) have improved the quality and the versatility of a character motion compared to other methods. Tracking-based learning approaches show realistic motions by tracking a reference motion clip. However, it becomes difficult to apply this technique for multiple disparate motion clips since it is challenging to select a suitable reference motion for the character to track at each time step. Also, imitation learning requires manually-designed pose error metrics to compute imitation rewards. It can be challenging to design and tune these metrics for better performances.

Generative adversarial imitation learning (GAIL) proposes a plausible solution for the scalability of imitation learning [Ho and Ermon 2016]. GAIL trains a policy with an adversarial discriminator which distinguishes between behaviors from an agent and behaviors from demonstration data. This algorithm enables a character to imitate multiple motion clips, as well as design rewards from the adversarial discriminator instead of manually-designed imitation rewards. Although characters can learn multiple motion skills using this method, designing an interactive character controller is another problem. Adversarial motion prior controls physics-based character using GAIL, but it requires manually-designed goal-conditioned rewards for each task [Peng et al. 2021]. Also, it is challenging to preserve high quality of motions while the character is exploring extensive motion space and learning numerous motor skills.

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SIGGRAPH '22 Posters, August 07-11, 2022, Vancouver, BC, Canada

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ACM ISBN 978-1-4503-9361-4/22/08.

<https://doi.org/10.1145/3532719.3543224>

In this paper, we present an algorithm that learns an diverse motion space from disparate motion clips. The character learns various motor skills and shows suitable actions following user commands at run-time. The character can punch and kick sequentially while walking or running. When a new user command is given, the character plans a motion sequence that performs suitable motor skills from the current pose without any motion planners.

2 OUR APPROACH

Our main algorithm is developed based on GAIL, in which a discriminator distinguish an agent's behavior in RL, and a policy in RL gets rewards from the discriminator. The goal of our research is to learn various dynamic motions without complex reward designs and change motion types following user control. Our approach for this goal is composed of 2 ideas: (1) Conditioning discriminators and (2) Reward design using discriminator outputs.

2.1 Conditioning Discriminator

Our discriminator takes inputs from both datasets and the character's behaviors and does two tasks. First, our discriminator determines whether a given motion is from demonstrations or an agent(character)'s behavior. Moreover, our discriminator classifies the given motion into a specific motion class. These tasks require our discriminator to evaluate the quality of the agent's behavior as well as assess how well the agent performs following user intentions.

To design our discriminator with conditions, the network architecture of our discriminator is derived from a discriminator with an auxiliary classifier(ACGAN) [Odena et al. 2017]. Therefore, the discriminator objective function is given by:

$$L = L_{style} + L_{class} + L_{gp} \quad (1)$$

where L_{style} , L_{class} , L_{gp} are style loss, classification loss, and gradient penalty loss, respectively. Style loss L_{style} forces the discriminator to distinguish between the agent's behavior and behaviors from demonstrations. Classification loss L_{class} is a cross-entropy loss between target classes and predicted classes from our discriminator. Target classes can be the user's control inputs or labeled classes on dataset. Gradient penalty L_{gp} is derived from the objective function in Adversarial motion prior [Peng et al. 2021]. This term is crucial for stable training and effective performances.

2.2 Reward design

Reward terms in RL are composed of 2 components: (1) style reward r^S and (2) control reward r^C .

$$r_t(s_t, a_t, s_{t+1}, \hat{c}_t) = w^S r^S(s_t, s_{t+1}) + w^C r^C(s_t, s_{t+1}, \hat{c}_t) \quad (2)$$

where r_t is a reward at each frame t , and $s_t, a_t, \hat{c}_t$ are the state, the action of an agent, and user control input at frame t , respectively. The two reward terms are combined linearly with each weight w^S and w^C . Since both two rewards are derived from the conditioned discriminator, there is no need to manually design rewards related to pose error metrics.

$$r^S(s_t, s_{t+1}) = \mathcal{F}(D_1(s_t, s_{t+1})) \quad (3)$$

$$r^C(s_t, s_{t+1}, \hat{c}_t) = \mathcal{G}(D_2(s_t, s_{t+1}), \hat{c}_t) \quad (4)$$

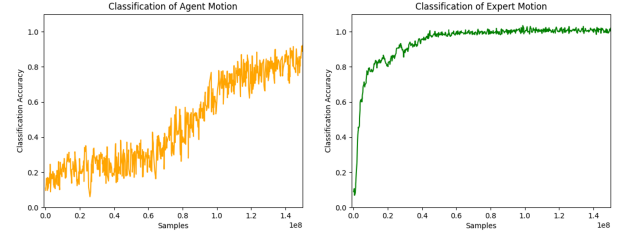


Figure 2: The classification accuracy of our discriminator about 10 kinds of motions. The orange line shows classification accuracy for agent behaviors, and the green line shows classification accuracy for expert(motion capture) data. We collected 150 million samples for training our policy and discriminator.

where D_1 and D_2 are discriminator outputs. $\mathcal{F}$ and $\mathcal{G}$ are manually designed functions. Function $\mathcal{F}$ enables a character to get a high reward when its behavior fools the discriminator well. Otherwise, $\mathcal{G}$ computes a high reward when the character's motion type predicted by the discriminator corresponds with user intention $\hat{c}_t$.

3 RESULT AND DISCUSSION

We designed a character controller which directs the character to perform 10 motor skills such as jump, kick, and breakdance. Our control policy is trained using proximal policy optimization(PPO). The detail of the character performance which shows a series of motions are shown in the supplementary video.

Figure 2 shows the classification accuracy of our conditional discriminator. While accuracy for expert data increases dramatically at the early stages, accuracy for agent behaviors shows a gradual increase with a smaller gradient than accuracy for expert data. From this result, we infer that our discriminator gradually classifies the agent behaviors well as our agent performs realistic motions like motion capture data. Also, we concluded that our discriminator has a high capability of learning diverse motor skills since the discriminator shows good performances in differentiating agent's behaviors from expert data as well as classifying the types of input motions.

Our future direction is to improve the motion quality for each motor skill and expands the range of motions that a character can learn. When various motor skills are given, characters are likely to learn easy motions and fail to try difficult motions. Also, network stability is always considered as the major issue in adversarial learning. If the network fails to keep stability, the motion quality will be degraded and a character will fail to learn dynamic motor skills.

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